THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 20	$E = 5F/2 = 30$; $V = 2 + E - F$
2	**(d)** 8	$V = 2E/3 = 12$; $F = 2 + E - V$
3	**(a)** 24	Each corner is replaced by 3 new vertices: $8 \times 3 = 24$
4	**(a)** 4	$6(n-2)^2 = 24$ gives $n = 4$
5	**(c)** 256	$81 \times (4/3)^4$
6	**(d)** 4	Side halves every stage: $64/2^4$
7	**(d)** 8	$E = (3 \times 8 + 4 \times 2)/2 = 16$; $V = 2 + E - F$
8	**(d)** 26	$E = 5F/2 = 40$; $V = 2 + E - F$
9	**(a)** Both A and R are true and R is the correct explanation of A.	A cube has $F = 6$, $V = 8$, $E = 12$ and $6 + 8 - 12 = 2$; R is the general result.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A triangular prism has 2 triangular and 3 rectangular faces, so A is true; R is true but about pyramids, so it does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$12(n-2) = 48$ gives $n = 6$ (1); total $= 6^3 = 216$ (1).	1+1
12	Each edge is shared by two faces: $E = (3 \times 8 + 4 \times 6)/2$ (1) $= 24$ (1).	1+1
13	Number $= 3^5 = 243$ (1); side $= 128/2^5 = 4$ cm (1).	1+1
14	$12(n-2) = 84$ gives $n = 9$ (1); total $= 9^3 = 729$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$E = 3F/2 = 48$ (1); $V = 2 + E - F = 18$ (1); $F + V - E = 32 + 18 - 48 = 2$ (1).	1+1+1
16	$12(n-2) = 144$ gives $n = 14$ (1); one face: $6(n-2)^2 = 6 \times 144 = 864$ (2).	1+2
17	$3^5 = 243 \le 500$ but $3^6 = 729 > 500$ (2); so stage 6 with 729 triangles (1).	2+1
	OR: $E = 3F/2 = 60$ (1); $V = 2 + E - F = 22$ (1); $F + V - E = 40 + 22 - 60 = 2$ (1).	1+1+1
18	$12(n-2) = 168$ gives $n = 16$ (1); one face: $6(n-2)^2 = 6 \times 196 = 1176$ (2).	1+2
	OR: $3^6 = 729 \le 1000$ but $3^7 = 2187 > 1000$ (2); so stage 7 with 2187 triangles (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $5 + 2 = 7$ (1). (ii) $2 \times 5 = 10$ (1). (iii) $3 \times 5 = 15$ (1). (iv) Prism: $F + V - E = 7 + 10 - 15 = 2$ (1). Pyramid: $F = 6$, $V = 6$, $E = 10$ and $6 + 6 - 10 = 2$ (1).	1+1+1+2
20	$F = 16$; $E = (3 \times 12 + 4 \times 4)/2 = 26$ (2); $V = 2 + E - F = 12$ (2); $F + V - E = 2$ (1).	2+2+1
	OR: Stage 0 perimeter $= 27$ cm (1); each stage multiplies the perimeter by $\frac{4}{3}$ (1); stage 2: $3 \times 9 \times (4/3)^2 = 48$ cm (2); increase $= 21$ cm (1).	1+1+2+1

Mistakes to avoid:

- Counting faces, edges and vertices of a prism and pyramid with each other's formulas (a pyramid has $2n$ edges, a prism $3n$).
- Forgetting the apex when counting the vertices of a pyramid (a square pyramid has 5, not 4).
- In painted-cube questions, using $n - 2$ for the edge cubes but forgetting to multiply by the number of edges (12) or faces (6).
- Assuming that any six squares joined edge to edge fold into a cube; the net must not overlap when folded.
- Using Euler's formula as $F + V + E = 2$; the correct form is $F + V - E = 2$.
- Counting the stages of a fractal wrongly: the starting figure is stage 0, not stage 1.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 5	$6(n-2)^2 = 54$ gives $n = 5$
2	**(c)** 64	$27 \times (4/3)^3$
3	**(c)** 12	Side halves every stage: $96/2^3$
4	**(d)** 6	$E = (3\times4 + 4\times2)/2 = 10$; $V = 2 + E - F$
5	**(d)** 32	$E = 5F/2 = 50$; $V = 2 + E - F$
6	**(d)** 14	$V = 2E/4 = 12$; $F = 2 + E - V$
7	**(a)** 36	$F + V - 2 = 14 + 24 - 2$
8	**(b)** 7	$6(n-2)^2 = 150$ gives $n = 7$
9	**(c)** A is true but R is false.	Edges $= 3 \times 6 = 18$ so A is true; a hexagonal prism has 12 vertices, so R is false.
10	**(d)** A is false but R is true.	A square pyramid has 5 vertices (4 base + 1 apex), so A is false; it has 8 edges, so R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Each edge is shared by two faces: $E = (3\times20 + 4\times0)/2$ (1) $= 30$ (1).	1+1
12	Number $= 3^5 = 243$ (1); side $= 96/2^5 = 3$ cm (1).	1+1
13	$12(n-2) = 96$ gives $n = 10$ (1); total $= 10^3 = 1000$ (1).	1+1
14	Each edge is shared by two faces: $E = (3\times12 + 4\times6)/2$ (1) $= 30$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$E = 3F/2 = 54$ (1); $V = 2 + E - F = 20$ (1); $F + V - E = 36 + 20 - 54 = 2$ (1).	1+1+1
16	$12(n-2) = 192$ gives $n = 18$ (1); one face: $6(n-2)^2 = 6 \times 256 = 1536$ (2).	1+2
17	$3^7 = 2187 \le 5000$ but $3^8 = 6561 > 5000$ (2); so stage 8 with 6561 triangles (1).	2+1
	OR: $E = 3F/2 = 66$ (1); $V = 2 + E - F = 24$ (1); $F + V - E = 44 + 24 - 66 = 2$ (1).	1+1+1
18	$12(n-2) = 228$ gives $n = 21$ (1); one face: $6(n-2)^2 = 6 \times 361 = 2166$ (2).	1+2
	OR: $3^8 = 6561 \le 10000$ but $3^9 = 19683 > 10000$ (2); so stage 9 with 19683 triangles (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 8 corner cubes (1). (ii) $12 \times (4-2) = 24$ (1). (iii) $6 \times (4-2)^2 = 24$ (1). (iv) $(4-2)^3 = 8$ (1); $8 + 24 + 24 + 8 = 64$ (1).	1+1+1+2
20	$F = 24$; $E = (3\times16 + 4\times8)/2 = 40$ (2); $V = 2 + E - F = 18$ (2); $F + V - E = 2$ (1).	2+2+1
	OR: Stage 0 perimeter $= 54$ cm (1); each stage multiplies the perimeter by $\frac{4}{3}$ (1); stage 2: $3 \times 18 \times (4/3)^2 = 96$ cm (2); increase $= 42$ cm (1).	1+1+2+1

Mistakes to avoid:

- Counting faces, edges and vertices of a prism and pyramid with each other's formulas (a pyramid has $2n$ edges, a prism $3n$).
- Forgetting the apex when counting the vertices of a pyramid (a square pyramid has 5, not 4).
- In painted-cube questions, using $n-2$ for the edge cubes but forgetting to multiply by the number of edges (12) or faces (6).
- Assuming that any six squares joined edge to edge fold into a cube; the net must not overlap when folded.
- Using Euler's formula as $F + V + E = 2$; the correct form is $F + V - E = 2$.
- Counting the stages of a fractal wrongly: the starting figure is stage 0, not stage 1.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 8	$E = (3\times6 + 4\times3)/2 = 15$; $V = 2 + E - F$
2	**(a)** 38	$E = 5F/2 = 60$; $V = 2 + E - F$
3	**(a)** 8	$V = 2E/4 = 6$; $F = 2 + E - V$
4	**(d)** 10	Remove 1 vertex and add 3 new ones
5	**(b)** 8	$6(n-2)^2 = 216$ gives $n = 8$
6	**(b)** 16	$9 \times (4/3)^2$
7	**(d)** 10	Side halves every stage: $160/2^4$
8	**(c)** 6	$E = (3\times2 + 4\times3)/2 = 9$; $V = 2 + E - F$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$F = 2 - 10 + 15 = 7$; R is Euler's formula rearranged.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	The unit cube at the centre is unpainted, so A is true; the 8 corner cubes have three painted faces so R is true; R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Number $= 3^4 = 81$ (1); side $= 160/2^4 = 10$ cm (1).	1+1
12	$12(n-2) = 120$ gives $n = 12$ (1); total $= 12^3 = 1728$ (1).	1+1
13	Each edge is shared by two faces: $E = (3\times6 + 4\times8)/2$ (1) $= 25$ (1).	1+1
14	Number $= 3^4 = 81$ (1); side $= 144/2^4 = 9$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$E = 3F/2 = 72$ (1); $V = 2 + E - F = 26$ (1); $F + V - E = 48 + 26 - 72 = 2$ (1).	1+1+1
16	$12(n-2) = 264$ gives $n = 24$ (1); one face: $6(n-2)^2 = 6\times484 = 2904$ (2).	1+2
17	$3^9 = 19683 \le 20000$ but $3^10 = 59049 > 20000$ (2); so stage 10 with 59049 triangles (1).	2+1
	OR: $E = 3F/2 = 78$ (1); $V = 2 + E - F = 28$ (1); $F + V - E = 52 + 28 - 78 = 2$ (1).	1+1+1
18	$12(n-2) = 336$ gives $n = 30$ (1); one face: $6(n-2)^2 = 6\times784 = 4704$ (2).	1+2
	OR: $3^10 = 59049 \le 100000$ but $3^11 = 177147 > 100000$ (2); so stage 11 with 177147 triangles (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 3 (1). (ii) $3\times3 = 9$ (1). (iii) 27 (1). (iv) Each stage multiplies the count by 3, so the number is 3^n (1); at stage 5: $3^5 = 243$ (1).	1+1+1+2
20	$F = 18$; $E = (3\times6 + 4\times12)/2 = 33$ (2); $V = 2 + E - F = 17$ (2); $F + V - E = 2$ (1).	2+2+1
	OR: Stage 0 perimeter $= 108$ cm (1); each stage multiplies the perimeter by $\frac{4}{3}$ (1); stage 2: $3\times36\times(4/3)^2 = 192$ cm (2); increase $= 84$ cm (1).	1+1+2+1

Mistakes to avoid:

- Counting faces, edges and vertices of a prism and pyramid with each other's formulas (a pyramid has $2n$ edges, a prism $3n$).
- Forgetting the apex when counting the vertices of a pyramid (a square pyramid has 5, not 4).
- In painted-cube questions, using $n-2$ for the edge cubes but forgetting to multiply by the number of edges (12) or faces (6).
- Assuming that any six squares joined edge to edge fold into a cube; the net must not overlap when folded.
- Using Euler's formula as $F + V + E = 2$; the correct form is $F + V - E = 2$.
- Counting the stages of a fractal wrongly: the starting figure is stage 0, not stage 1.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.